

In this project, you will be implementing and creating your own **Voronoi Diagram**.

TASKS

For the final project, you will be tasked to create a Voronoi diagram using any combination of algorithms and operations that we have discussed in class. A Voronoi diagram, as discussed, is composed of several points called sites divided into regions or cells that are bounded by edges.

You are allowed to use **any** algorithm that we have discussed in class. These are:

1. **Perpendicular Bisectors**
2. **Fortune's Algorithm**
3. **Delaunay Triangulation / Bowyer-Watson's Algorithm**

You are allowed to use **ANY** of the previous programming exercise ZIP files as your starting code for your project. This also means that you are free to make your own test cases and change the rest of the codes within the programming exercises for you to render the Voronoi diagram.

OUTPUT

Your outputs (rendered programs) should have the following specifications:

1. Window size: 1280x720.
2. Window name: LastNameLastNameLastname - Voronoi Diagram
3. All **SITES** must be rendered/colored **white**.
4. All **REGIONS** must be **uniquely colored**.
5. All **EDGES** are rendered/colored in **red**.

There are also additional bonus points that you can garner alongside the base specifications. You can get a **total up to 10 bonus points** for implementing any number of the following:

1. (5pts) Implement a "player" that could be controlled with the keyboard that highlights the region the "player" is in and the closest site from the "player"
2. (10pts) Animate the creation of the Voronoi diagram using Fortune's algorithm.
3. (5pts) Render the Delaunay triangulation on top of the Voronoi diagram. The triangulation must disappear and reappear on a key press.
4. **FREESTYLE BONUS** - Implement any additional features or mechanics for a variable amount of points.

SUBMISSION, DEADLINE, AND DEFENSES

Submit your **Main.cpp** file along with a filled-in **Certificate of Authorship** to the submission link in our class website at Canvas (Final Project Submission). After submitting, sign-up for a **code defense**. The sign-up sheet will be available at the course's Canvas page. Defenses will be held onsite at **F-206 (ALLS)**.

The deadline for submitting your final project is November 26, 2025.